Networked & Distributed Control: Assignment 3

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***Abstract*—This report presents the step-by-step solution to Assignment 3 of the SC42101 course (Networked and Distributed Control Systems). The assignment explores optimality conditions in convex quadratic programming, an iterative block-coordinate descent algorithm, its convergence behavior, and cost reduction analysis. Each problem is tackled analytically with an emphasis on mathematical rigor and interpretation.**

I. Introduction

The aim of this assignment is to develop deeper understanding of convex optimization in control, particularly focusing on coordinate descent algorithms and their properties such as convergence, optimality, and cost monotonicity. Each question addresses one of these properties, and we use matrix algebra, convex analysis, and coordinate transformations to prove the required results.

II. Problem 1: First-order Optimality and Normal Cone

A convex quadratic problem is given by:

$$\min_{x\in X} \quad f(x) = \frac{1}{2}x^\top Hx + c^\top x,$$

where $H \geq 0$ and X is a convex set.

A. Goal:

Show that x^* is optimal if and only if:

$$(z - x^*)^\top(-\nabla f(x^*)) \leq 0, \quad \forall z \in X.$$

B. Solution:

Differentiating the convex quadratic function gives::

$$\nabla f(x) = Hx + c$$

and so this also holds for x^*:

$$\nabla f(x^*) = Hx^* + c$$

Here, x^* is only optimal if the directional derivative towards the optimal point is non-negative. This results in the following first optimality condition:

$$\nabla f(x^*)^\top(z - x^*) \geq 0 \iff (z - x^*)^\top(-\nabla f(x^*)) \leq 0.$$

Where the right hand side results from flipping the inequality.

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C. Normal Cone Interpretation:

The condition leads to the definition of the normal cone:

$$\mathcal{N}(X, x^*) = \left\{ y \mid (z - x^*)^\top (y - x^*) \leq 0, \forall z \in X \right\},$$

so that:

$$x^* \text{ is optimal} \iff -\nabla f(x^*) \in \mathcal{N}(X, x^*).$$

III. Problem 2: Iterative Block Optimization

Consider:

$$V(u) = \frac{1}{2}u^\top Hu + c^\top u + d, \quad u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

(a) Derivation of Update Rule

Optimize one block at a time:

$$u_1^{p+}(u_2^p) = -H_{11}^{-1}(H_{12}u_2^p + c_1)$$
$$u_2^{p+}(u_1^p) = -H_{22}^{-1}(H_{21}u_1^p + c_2)$$

These can be combined in vector u^{p+1}:

$$u^{p+1} = \begin{bmatrix} u_1^{p+1} \\ u_2^{p+1} \end{bmatrix} = w_1 \begin{bmatrix} u_1^{p+} \\ u_2^p \end{bmatrix} + w_2 \begin{bmatrix} u_1^p \\ u_2^{p+} \end{bmatrix}$$

Then, filling in the values for u^{p+} into the previous equation. This can be written in the form $u^{p+1} = A * u^p + b$ as follows:

$$u^{p+1} = w_1 \begin{bmatrix} -H_{11}^{-1}(H_{12}u_2^p + c_1) \\ u_2^p \end{bmatrix} + w_2 \begin{bmatrix} u_1^p \\ -H_{22}^{-1}(H_{21}u_1^p + c_2) \end{bmatrix}$$

$$= \begin{bmatrix} w_2 I & -w_1 H_{11}^{-1}H_{12} \\ -w_2 H_{22}^{-1}H_{21} & w_1 I \end{bmatrix} \begin{bmatrix} u_1^p \\ u_2^p \end{bmatrix} + \begin{bmatrix} -w_1 H_{11}^{-1}c_1 \\ -w_2 H_{22}^{-1}c_2 \end{bmatrix}$$

Resulting in the following values for A and b:

$$A = \begin{bmatrix} w_2 I & -w_1 H_{11}^{-1}H_{12} \\ -w_2 H_{22}^{-1}H_{21} & w_1 I \end{bmatrix}, \quad b = \begin{bmatrix} -w_1 H_{11}^{-1}c_1 \\ -w_2 H_{22}^{-1}c_2 \end{bmatrix}.$$

(b) Convergence of the Iteration

Given $H \succ 0$ and $w_1 + w_2 = 1$, the iteration matrix A is stable:

$$|\lambda_i(A)| < 1 \quad \text{(spectral radius)}.$$

(c) Convergence to Global Optimum

This is the differential rewritten for the optimal point. Giving the following:

$$\nabla V(u) = 0 \iff \nabla V(u^*) = 0$$

At steady-state:

$$u^* = Au^* + b \Rightarrow (I - A)u^* = b \Rightarrow u^* = -H^{-1}c$$

This is the differential rewritten for the optimal point.

IV. Problem 3: Monotonic Decrease of Cost

We show that:

$$V(u^{p+1}) < V(u^p), \quad \forall u^p \neq u^* = -H^{-1}c.$$

With:

$$V(u^{p+1}) - V(u^p) = -\frac{1}{2}(u^p - u^*)^\top P(u^p - u^*)$$

Where:

$$P = HD^{-1}\tilde{H}D^{-1}H, \quad \tilde{H} = D - N, \quad H = D + N$$

$$D = \begin{bmatrix} w_1^{-1}H_{11} & 0 \\ 0 & w_2^{-1}H_{22} \end{bmatrix}, \quad N = \begin{bmatrix} -w_1^{-1}w_2H_{11} & H_{12} \\ H_{21} & -w_2^{-1}w_1H_{22} \end{bmatrix}$$

The result follows from a coordinate shift $\tilde{u} = u - u^*$ and update rule $u^{p+1} = Au^p + b$.

V. Problem 4: 3-Block Coordinate Descent

Perform 3-step updates:

$$u_1^{p+1} = \arg\min_{u_1} V(u_1, u_2^p, u_3^p)$$

$$u_2^{p+1} = \arg\min_{u_2} V(u_1^{p+1}, u_2, u_3^p)$$

$$u_3^{p+1} = \arg\min_{u_3} V(u_1^{p+1}, u_2^{p+1}, u_3)$$

Performing each update separately and in order will result in:

$$V(u_1^{p+1}, u_2^p, u_3^p) \leq V(u_1^p, u_2^p, u_3^p)$$

Filling this step into the next one gives a similar result:

$$V(u_1^{p+1}, u_2^{p+1}, u_3^p) \leq V(u_1^{p+1}, u_2^p, u_3^p)$$
$$V(u_1^{p+1}, u_2^{p+1}, u_3^{p+1}) \leq V(u_1^{p+1}, u_2^{p+1}, u_3^p)$$

Each step reduces or maintains the cost due to convexity, more generally resulting into:

$$V(u^{p+1}) \leq V(u^p)$$

D. Example

We fix $u_2 = 0$ and $u_3 = 1$, and minimize $V(u)$ over u_1:

$$u = \begin{bmatrix} u_1 \\ 0 \\ 1 \end{bmatrix}$$

Compute:

$$Hu = \begin{bmatrix} 2u_1 + 1 \\ u_1 + 1 \\ u_1 + 2 \end{bmatrix}, \quad u^\top Hu = 2u_1^2 + 2u_1 + 2, \quad c^\top u = 1$$

$$V(u) = \frac{1}{2}u^\top Hu + c^\top u = \frac{1}{2}(2u_1^2 + 2u_1 + 2) + 1 = u_1^2 + u_1 + 2$$

Take derivative:

$$\frac{dV}{du_1} = 2u_1 + 1 = 0 \Rightarrow u_1^{p+1} = -\frac{1}{2}$$

So the updated intermediate vector becomes:

$$u^{p,\text{intermediate}} = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$$

Which shows the distance is decreasing.

VI. Conclusion

This report has established foundational optimality conditions in convex programming, derived and analyzed a block-coordinate descent method, and demonstrated its convergence and descent properties. These insights are valuable for decentralized or distributed control strategies.

References